

Problem Set 4 Solutions

QTM 220

Due 20 October at 2:00pm ET on Canvas. Questions about submission requirements should be directed to your TA.

NBA Draft Data: Sample

The data set `nba.sample.draft` includes the following variables from the 2021-2022 NBA season:

- `playername`: name of player
- `teamabbreviation` : name of the team the player played on at the end of the season
- `age`: age of player
- `playerheight` : height in cm
- `playerweight`: weight in kg
- `college`: name of college player attended
- `country` : name of country where player was born
- `draftyear` : year player was drafted ('Undrafted' if undrafted)
- `drafteround` : draft round player was selected
- `draftnumber` : number selected in his draft round ('Undrafted' if undrafted)
- `gp`: games played through season
- `pts`: average points scored
- `reb`: average rebounds
- `ast`: average assists
- `netrating` : team's point differential per 100 possessions while player is on the court

- orebpct : percentage of available offensive rebounds grabbed
- drebpct: percentage of available defensive rebounds grabbed
- usgpct: percentage of team plays used $(FGA + \text{Possession Ending FTA} + TO)/POSS$
- tspct: shooting efficiency $(PTS/(2*(FGA + 0.44*FTA)))$
- astpct : percent of teammate field goals assisted
- draftlevel : draft round player was selected with undrafted players coded as 3

You are interested in the average net rating, the team's point differential per 100 possessions while the player is on the court, among different draft levels– players drafted in the first round, second round, or undrafted.

- (a) What is the subsample mean net rating among those drafted in the first round? Second round? Undrafted? Create a scatterplot with draft level on the x-axis and net rating on the y-axis.

```
mean(nba.sam$net_rating[nba.sam$draftlevel==1])
```

```
[1] 0.7130435
```

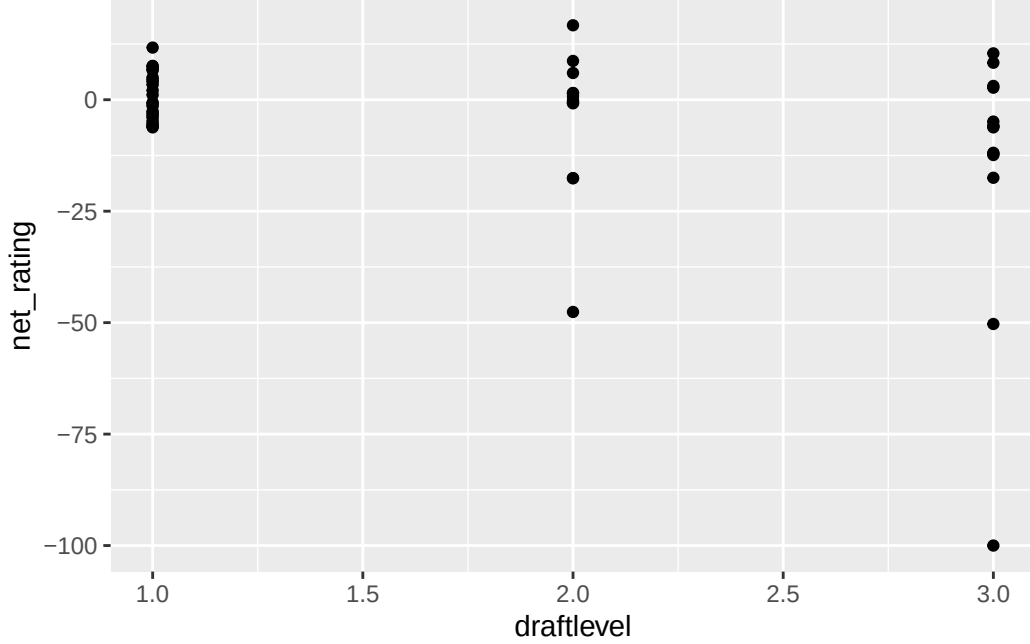
```
mean(nba.sam$net_rating[nba.sam$draftlevel==2])
```

```
[1] -3.823077
```

```
mean(nba.sam$net_rating[nba.sam$draftlevel==3])
```

```
[1] -14.47143
```

```
ggplot(nba.sam, aes(x=draftlevel, y=net_rating)) +geom_point ()
```



- (b) Using the notation from class, show that the subsample mean net rating estimator among those drafted in the first round is unbiased for the subpopulation mean net rating among those drafted in the first round.

We want to show that $E[\hat{\mu}(1)] = \mu(1)$

Subsample mean net rating estimator: $\hat{\mu}(1) = \frac{\sum_{i=1}^n 1_1(X_i)Y_i}{\sum_{i=1}^n 1_1(X_i)}$ where

$$1_1(X) = \begin{cases} 1 & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

By law of iterated expectations, $E[\hat{\mu}(1)] = E[E[\frac{\sum_{i=1}^n 1_1(X_i)Y_i}{\sum_{i=1}^n 1_1(X_i)} | X_1, \dots, X_n]]$

$$E[\frac{1}{\sum_{i=1}^n 1_1(X_i)} E[\sum_{i=1}^n 1_1(X_i)Y_i | X_1, \dots, X_n]]$$

because when we condition the indicator sum is a constant

$$E[\frac{1}{\sum_{i=1}^n 1_1(X_i)} \sum_{i=1}^n E[1_1(X_i)Y_i | X_1, \dots, X_n]]$$

expectation of sum = sum of expectations

$$E[\frac{1}{\sum_{i=1}^n 1_1(X_i)} \sum_{i=1}^n 1_1(X_i) E[Y_i | X_1, \dots, X_n]]$$

$$E\left[\frac{1}{\sum_{i=1}^n 1_1(X_i)} \sum_{i=1}^n 1_1(X_i) E[Y_i|X_i]\right]$$

by irrelevance of independent conditioning variables

We know $E[Y_i|X_i] = \mu(X_i)$ by definition

$$E\left[\frac{1}{\sum_{i=1}^n 1_1(X_i)} \sum_{i=1}^n 1_1(X_i) \mu(X_i)\right]$$

for all $\mu(X_i)$ that are not $\mu(1)$ (i.e. $\mu(2)$ and $\mu(3)$), the indicator function will multiply them by 0 so we can remove them

$$E\left[\frac{1}{\sum_{i=1}^n 1_1(X_i)} \sum_{i=1}^n 1_1(X_i) \mu(1)\right]$$

$\mu(1)$ is a constant

$$E\left[\frac{1}{\sum_{i=1}^n 1_1(X_i)} \mu(1) \sum_{i=1}^n 1_1(X_i)\right]$$

The sums cancel

$$E[\mu(1)] = \mu(1)$$

as expectation of a constant is that constant

Thus the subsample mean among the first round draft picks is an unbiased estimator of the subpopulation mean among first round draft picks.

- (c) What is the average incremental difference in mean net rating across the draft levels in your sample using the draftlevel variable,

$$\frac{1}{2} \sum_{x=2}^3 \{\hat{\mu}(x) - \hat{\mu}(x-1)\}$$

```
(1/2)*((mean(nba.sam$net_rating[nba.sam$draftlevel==2]) -
        mean(nba.sam$net_rating[nba.sam$draftlevel==1])) +
        (mean(nba.sam$net_rating[nba.sam$draftlevel==3]) -
         mean(nba.sam$net_rating[nba.sam$draftlevel==2])))
```

```
[1] -7.592236
```

```
(1/2)*(mean(nba.sam$net_rating[nba.sam$draftlevel==3]) -
        mean(nba.sam$net_rating[nba.sam$draftlevel==1]))
```

```
[1] -7.592236
```

- (d) Rewrite part (c) as a linear combination of sub-sample means, $\sum_x \alpha(x) \hat{\mu}(x)$. What is your $\alpha(x)$ function?

$\sum_{i=1}^3 \alpha(x) \hat{\mu}(x)$ where

$$\alpha(x) = \begin{cases} -\frac{1}{2} & \text{if } x = 1 \\ \frac{1}{2} & \text{if } x = 3 \\ 0 & \text{otherwise} \end{cases}$$

This yields $\frac{1}{2} \hat{\mu}(3) - \frac{1}{2} \hat{\mu}(1)$

```
(1/2)*mean(nba.sam$net_rating[nba.sam$draftlevel==3]) - (1/2)*mean(nba.sam$net_rating[nba.s
```

```
[1] -7.592236
```

- (e) What is the estimated standard error of your average incremental difference using plug-in/normal approximation?

The variance of our estimator is $\sum_x \frac{\alpha^2(x) \hat{\sigma}^2(x)}{N_x}$, which we can take the square root of for the estimated standard error. First, lets calculate the sub-sample variances.

```
v1<- var(nba.sam$net_rating[nba.sam$draftlevel==1])
v2<- var(nba.sam$net_rating[nba.sam$draftlevel==2])
v3<- var(nba.sam$net_rating[nba.sam$draftlevel==3])
```

We know our α s are $\frac{1}{2}$ and $-\frac{1}{2}$

What are our sub-sample sizes?

```
n1<- length(nba.sam$net_rating[nba.sam$draftlevel==1])
n2<- length(nba.sam$net_rating[nba.sam$draftlevel==2])
n3<- length(nba.sam$net_rating[nba.sam$draftlevel==3])
```

Putting it all together,

```
((-1/2)^2)*(v1))/n1
```

```
[1] 0.3042915
```

```
((1/2)^2)*(v3))/n3
```

```
[1] 14.65965
```

```
(((-1/2)^2)*(v1))/n1 + ((1/2)^2)*(v3))/n3
```

```
[1] 14.96395
```

That is the variance of our estimator, but we need the standard error, which is the square root

```
sqrt((((-1/2)^2)*(v1))/n1 + ((1/2)^2)*(v3))/n3)
```

```
[1] 3.868326
```

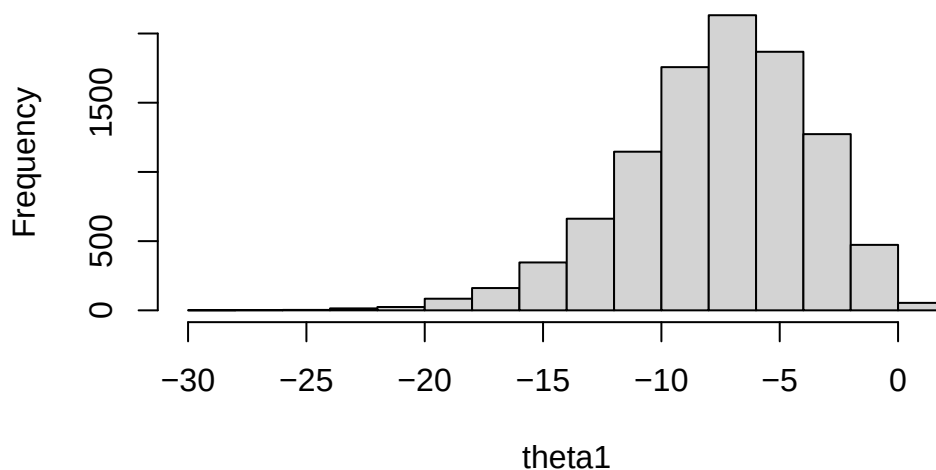
- (f) Generate a bootstrap estimated sampling distribution of the average incremental difference estimator. Calculate a 95% confidence interval using your bootstrapped distribution.

```
set.seed(220)
B=10000
theta1<- rep(NA,B)
n=50

for( b in 1:B ){
  bootind = sample(1:n,size=n,replace=T)
  dfboot = nba.sam[bootind,]
  theta1[b] = (1/2)*mean(dfboot$net_rating[dfboot$draftlevel==3]) -(1/2)*mean(dfboot$net_rat
}

hist(theta1)
```

Histogram of theta1



```
sd(theta1)
```

```
[1] 3.891685
```

```
quantile(theta1,prob =c(.025,.975))
```

```
      2.5%      97.5%
-16.381284 -1.274904
```

- (g) What is the average incremental difference in mean net rating across the draft levels in your sample weighted by the proportion of players drafted in each round (and undrafted)?

$$\sum_{x=2}^3 P_x \{ \hat{\mu}(x) - \hat{\mu}(x-1) \}$$

$$\text{where } P_x = \frac{N_x}{\sum_{x=2}^3 N_x}$$

Our P s: $P_2 = \frac{13}{13+14}$; $P_3 = \frac{14}{13+14}$

$$P_2(\hat{\mu}(2) - \hat{\mu}(1)) + P_3(\hat{\mu}(3) - \hat{\mu}(2))$$

```
(13/27)*(mean(nba.sam$net_rating[nba.sam$draftlevel==2]) - mean(nba.sam$net_rating[nba.sam
```

[1] -7.705425

- (h) Rewrite part (g) as a linear combination of sub-sample means, $\sum_x \hat{\alpha}(x)\hat{\mu}(x)$. What is your $\hat{\alpha}(x)$ function?

$$P_2\hat{\mu}(2) - P_2\hat{\mu}(1) + P_3\hat{\mu}(3) - P_3\hat{\mu}(2)$$

$$\sum_{i=1}^3 \hat{\alpha}(x)\hat{\mu}(x) \text{ where}$$

$$\hat{\alpha}(x) = \begin{cases} -P_2 & \text{if } x = 1 \\ P_2 - P_3 & \text{if } x = 2 \\ P_3 & \text{if } x = 3 \end{cases}$$

- (i) What is the lower bound on the estimated standard error of your sub-sample weighted average incremental difference using plug-in/normal approximation?

We already have our subsample variances and subsample sizes from earlier, but now our α s are different

$$((-13/27)^2 * v1)/n1$$

[1] 0.2821688

$$((13/27 - 14/27)^2 * v2)/n2$$

[1] 0.027192

$$((14/27)^2 * v3)/n3$$

[1] 15.76566

$$((-13/27)^2 * v1)/n1 + ((13/27 - 14/27)^2 * v2)/n2 + ((14/27)^2 * v3)/n3$$

[1] 16.07503

$$\text{sqrt}(((13/27)^2 * v1)/n1 + ((13/27 - 14/27)^2 * v2)/n2 + ((14/27)^2 * v3)/n3)$$

```
[1] 4.009367
```

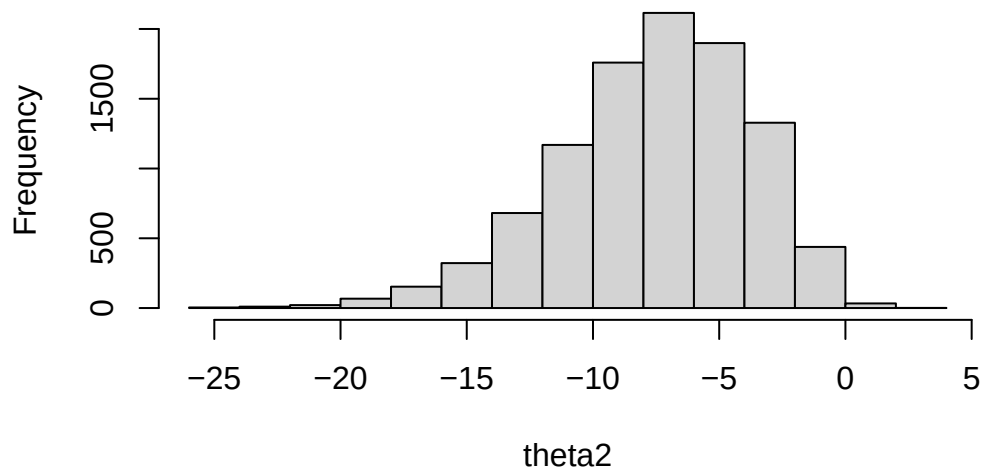
- (j) Generate a bootstrap estimated sampling distribution of the weighted average incremental difference estimator. Calculate a 95% confidence interval using your bootstrapped distribution.

```
theta2<- rep(NA,B)

for( b in 1:B ){
  bootind2 = sample(1:n,size=n,replace=T)
  dfboot2 = nba.sam[bootind2,]
  P2<- length(dfboot2$draftlevel==2)/(length(dfboot2$draftlevel==2) + length(dfboot2$draftle
  P3<- length(dfboot2$draftlevel==3)/(length(dfboot2$draftlevel==2) + length(dfboot2$draftle
  theta2[b] = (-P2)*mean(dfboot2$net_rating[dfboot2$draftlevel==1]) + (P2-P3)*mean(dfboot2$m
}

hist(theta2)
```

Histogram of theta2



```
sd(theta2)
```

```
[1] 3.79976
```

```
quantile(theta2,prob =c(.025,.975))
```

2.5%	97.5%
-16.036839	-1.396534

(k) What is the least squares difference in mean net rating across draft levels in your sample?

$$\sum_x \tilde{w}(x) \hat{\mu}(x)$$

$$\text{where } \tilde{w}(x) = \frac{N_x(x-\bar{X})}{\sum_i (X_i - \bar{X})^2}$$

```
xvar<- rep(NA, n)
for(i in 1:n){
  xvar[i]<- (nba.sam$draftlevel[i] - mean(nba.sam$draftlevel))^2
}
denom<- sum(xvar)
```

$$\tilde{w}(1) = \frac{23*(1-\bar{X})}{\sum_i (X_i - \bar{X})^2}$$

$$\tilde{w}(2) = \frac{13*(2-\bar{X})}{\sum_i (X_i - \bar{X})^2}$$

$$\tilde{w}(3) = \frac{14*(3-\bar{X})}{\sum_i (X_i - \bar{X})^2}$$

```
((23*(1-mean(nba.sam$draftlevel)))/denom)*mean(nba.sam$net_rating[nba.sam$draftlevel==1])
```

```
[1] -7.390107
```

```
#Just to check because it should be the same as the slope:
mod<- lm(net_rating~draftlevel, data=nba.sam)
summary(mod)
```

Call:

```
lm(formula = net_rating ~ draftlevel, data = nba.sam)
```

Residuals:

Min	1Q	Median	3Q	Max
-86.562	-4.272	2.958	7.463	23.838

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.732	5.852	1.492	0.1422
draftlevel	-7.390	2.919	-2.532	0.0147 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 17.36 on 48 degrees of freedom

Multiple R-squared: 0.1178, Adjusted R-squared: 0.09943

F-statistic: 6.41 on 1 and 48 DF, p-value: 0.01468

- (l) Rewrite part (k) as a linear combination of sub-sample means using the $\alpha(x)$ notation, $\sum_x \hat{\alpha}(x)\hat{\mu}(x)$. What is your $\hat{\alpha}(x)$ function?

$\hat{\alpha}(x)$ is the same as our $\tilde{w}(x)$ s

$\sum_{i=1}^3 \hat{\alpha}(x)\hat{\mu}(x)$ where

$$\hat{\alpha}(x) = \begin{cases} \frac{N_1*(1-\bar{X})}{\sum_i (X_i - \bar{X})^2} & \text{if } x = 1 \\ \frac{N_2*(2-\bar{X})}{\sum_i (X_i - \bar{X})^2} & \text{if } x = 2 \\ \frac{N_3*(3-\bar{X})}{\sum_i (X_i - \bar{X})^2} & \text{if } x = 3 \end{cases}$$

- (m) What is the lower bound on the estimated standard error or your least squares summary of net rating across draft levels in your sample using plug-in/normal approximation?

We already have our subsample variances and subsample sizes from earlier, but now our α s are different

```
((23*(1-mean(nba.sam$draftlevel)))/denom))
```

```
[1] -0.5330695
```

```
((13*(2-mean(nba.sam$draftlevel)))/denom))
```

```
[1] 0.06613906
```

```
((14*(3-mean(nba.sam$draftlevel)))/denom))
```

```
[1] 0.4669305
```

```
((((23*(1-mean(nba.sam$draftlevel)))/denom))^2 *v1)/n1
```

```
[1] 0.3458737
```

```
((((13*(2-mean(nba.sam$draftlevel)))/denom))^2 * v2)/n2
```

```
[1] 0.08671311
```

```
((((14*(3-mean(nba.sam$draftlevel)))/denom))^2 *v3)/n3
```

```
[1] 12.78463
```

```
((((23*(1-mean(nba.sam$draftlevel)))/denom))^2 *v1)/n1 + (((13*(2-mean(nba.sam$draftlevel)))/denom))^2 *v2)/n2 + (((14*(3-mean(nba.sam$draftlevel)))/denom))^2 *v3)/n3
```

```
[1] 13.21722
```

```
sqrt((((23*(1-mean(nba.sam$draftlevel)))/denom))^2 *v1)/n1 + (((13*(2-mean(nba.sam$draftlevel)))/denom))^2 *v2)/n2 + (((14*(3-mean(nba.sam$draftlevel)))/denom))^2 *v3)/n3
```

```
[1] 3.635549
```

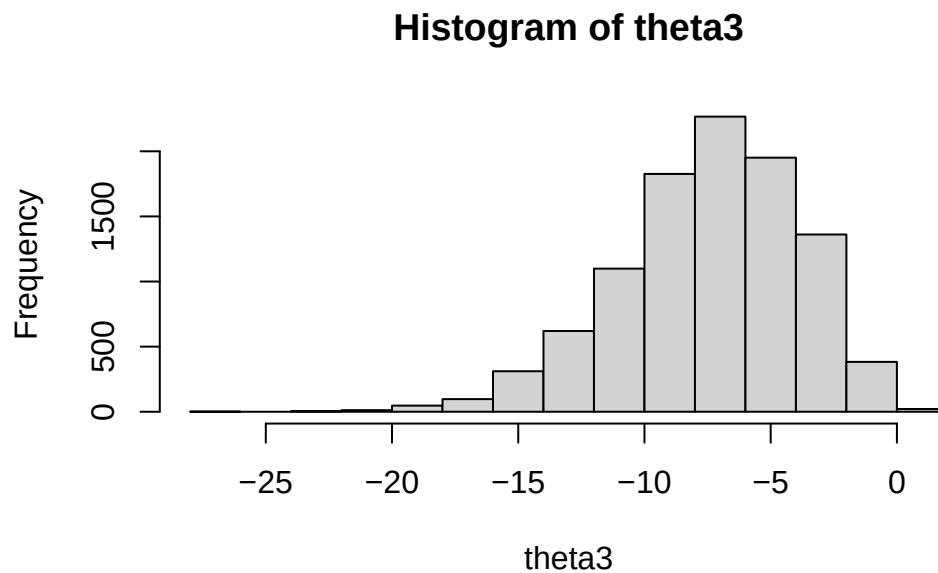
- (n) Generate a bootstrap estimated sampling distribution of the least squares estimator of net rating across draft levels. Calculate a 95% confidence interval using your bootstrapped distribution.

```
#this is a shortcut-- it would probably be better to calculate each of the weights and do

theta3<- rep(NA,B)

for( b in 1:B ){
bootind3 = sample(1:n,size=n,replace=T)
dfboot3 = nba.sam[bootind3,]
ols<- lm(dfboot3$net_rating~ dfboot3$draftlevel)
theta3[b] = ols$coef[2]
}
```

```
hist(theta3)
```



```
sd(theta3)
```

```
[1] 3.570151
```

```
quantile(theta3,prob =c(.025,.975))
```

2.5%	97.5%
-15.286754	-1.498534

(o) Fill in the following table with information using your sub-sample estimates

	x	1	2	3
	N_x	23	13	14
	$\hat{\mu}(x)$	0.713	-3.82	-14.47
	$\hat{\sigma}^2(x)$	21.995	257.699	820.941
avg. increment	$\alpha(x)$	$-\frac{1}{2}$	0	$\frac{1}{2}$
	$\alpha^2(x)$	$\frac{1}{4}$	0	$\frac{1}{4}$
	$\frac{\hat{\sigma}^2(x)\alpha^2(x)}{N_x}$	0.304	0	14.66
weighted increment	$\hat{\alpha}(x)$	$-\frac{13}{27}$	$\frac{13}{27} - \frac{14}{27}$	$\frac{14}{27}$
	$\hat{\alpha}^2(x)$	0.2318	0.00137	0.269
	$\frac{\hat{\sigma}^2(x)\hat{\alpha}^2(x)}{N_x}$	0.2821	0.02719	15.7657
least squares	$\hat{\alpha}(x)$	-0.533	0.06614	0.4669
	$\hat{\alpha}^2(x)$	0.284	0.004	0.2180
	$\frac{\hat{\sigma}^2(x)\hat{\alpha}^2(x)}{N_x}$	0.3459	0.0867	12.7846

Population Data

The data set nba_population includes the following variables from the 2021-2022 NBA season:

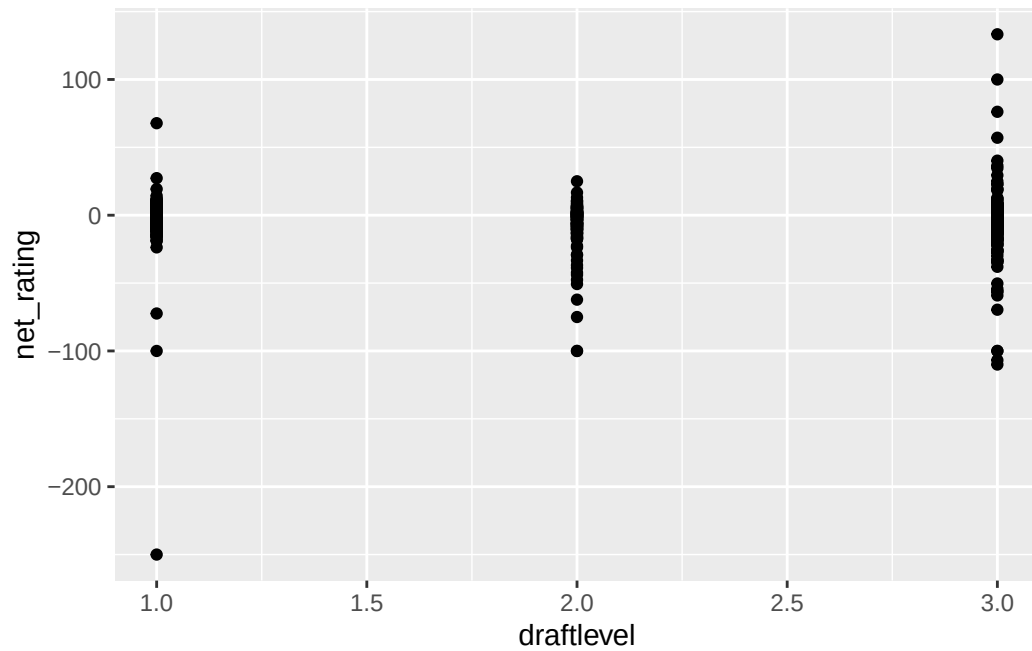
- playername: name of player
- teamabbreviation : name of the team the player played on at the end of the season
- age: age of player
- playerheight : height in cm
- playerweight: weight in kg

- college: name of college player attended
- country : name of country where player was born
- draftyear : year player was drafted ('Undrafted' if undrafted)
- draftround : draft round player was selected
- draftnumber : number selected in his draft round ('Undrafted' if undrafted)
- gp: games played through season
- pts: average points scored
- reb: average rebounds
- ast: average assists
- netrating : team's point differential per 100 possessions while player is on the court
- orebpc : percentage of available offensive rebounds grabbed
- drebpc: percentage of available defensive rebounds grabbed
- usgpct: percentage of team plays used (FGA + Possession Ending FTA + TO)/POSS
- tspct: shooting efficiency (PTS/(2*(FGA + 0.44*FTA)))
- astpct : percent of teammate field goals assisted
- draftlevel : draft round player was selected with undrafted players coded as 3

- (a) What are the sub-population mean net ratings among those drafted in the first round, second round, and undrafted? How do the sub-population means compare to your sub-sample means? How many players in the population are in each draft level? What is $\sigma^2(x)$ for each draft level? Fill in the table to organize your results.

x	1	2	3
n_x	279	144	182
$\mu(x)$	-1.4279	-7.0729	-5.4555
$\sigma^2(x)$	337.0019	315.3823	693.7344

```
ggplot(nba.pop, aes(x=draftlevel, y=net_rating)) +geom_point ()
```



```
np1<- length(nba.pop$net_rating[nba.pop$draftlevel==1])  
np1
```

```
[1] 279
```

```
np2<- length(nba.pop$net_rating[nba.pop$draftlevel==2])  
np2
```

```
[1] 144
```

```
np3<- length(nba.pop$net_rating[nba.pop$draftlevel==3])  
np3
```

```
[1] 182
```

```
np1+np2+np3
```

```
[1] 605
```

```
mu1<- mean(nba.pop$net_rating[nba.pop$draftlevel==1])
mu1
```

```
[1] -1.427957
```

```
mu2<- mean(nba.pop$net_rating[nba.pop$draftlevel==2])
mu2
```

```
[1] -7.072917
```

```
mu3<- mean(nba.pop$net_rating[nba.pop$draftlevel==3])
mu3
```

```
[1] -5.455495
```

```
vp1<- var(nba.pop$net_rating[nba.pop$draftlevel==1])
vp1
```

```
[1] 337.0016
```

```
vp2<- var(nba.pop$net_rating[nba.pop$draftlevel==2])
vp2
```

```
[1] 315.3823
```

```
vp3<- var(nba.pop$net_rating[nba.pop$draftlevel==3])
vp3
```

```
[1] 693.7344
```

The subsample means are quite different. The first round draft picks net rating is negative, on average, when it was positive in our sample. Our subsample mean of net ratings for undrafted players was also a lot lower than the population average– when we see the variance among undrafted players, it makes sense that our sample could be pretty off (it is also pretty small).

- (b) What is the average incremental difference in mean net rating across the draft levels in the population using the draftlevel variable,

$$\frac{1}{2} \sum_{x=2}^3 \{\mu(x) - \mu(x-1)\}$$

How does this compare to your estimate?

```
(1/2)*(mu3 - mu1)
```

```
[1] -2.013769
```

It is higher (closer to 0) than the sample. Our 95% confidence interval from the bootstrap covers.

- (c) Create a sampling distribution of your average incremental difference estimator. What size (n) should your repeated samples from the population be? What is the standard error for this estimator? Visually compare (with histograms) the sampling distribution to your bootstrap estimated sampling distribution for this estimator.

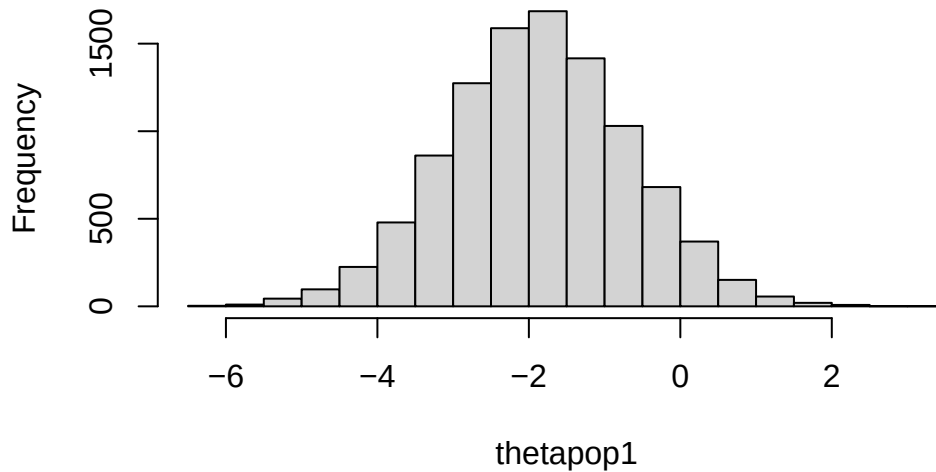
The repeated sample should be of size 50 (the same size of the sample data), but the samples are taken from the population.

```
S=10000
thetapop1<- rep(NA,S)
n=50

for( s in 1:S ){
  samind = sample(1:n,size=n,replace=T)
  dfsam = nba.pop[samind,]
  thetapop1[s] = (1/2)*mean(dfsam$net_rating[dfsam$draftlevel==3]) -(1/2)*mean(dfsam$net_rat
}

hist(thetapop1)
```

Histogram of thetapop1



```
sd(thetapop1)
```

```
[1] 1.203386
```

The standard error is smaller than our estimated standard error. The sampling distribution looks a lot more normal than the bootstrap estimated sampling distribution (maybe our sample is a bit small)

- (d) What is the average incremental difference in mean net rating across the draft levels in the population weighted by the proportion of players drafted in each round (and undrafted)?

$$\sum_{x=2}^3 p_x \{ \mu(x) - \mu(x-1) \}$$

where $p_x = \frac{m_x}{\sum_{x=2}^3 m_x}$ How does this compare to your estimate?

What are our population p s? $p_2 = \frac{144}{326}$; $P_3 = \frac{182}{326}$

$$p_2(\mu(2) - \mu(1)) + p_3(\mu(3) - \mu(2))$$

```
(np2/(np2+np3))*(mean(nba.pop$net_rating[nba.pop$draftlevel==2]) - mean(nba.pop$net_rating
```

```
[1] -1.590501
```

The true weighted increment average is higher than the sample, the 95% confidence interval we calculated before covers this true estimand.

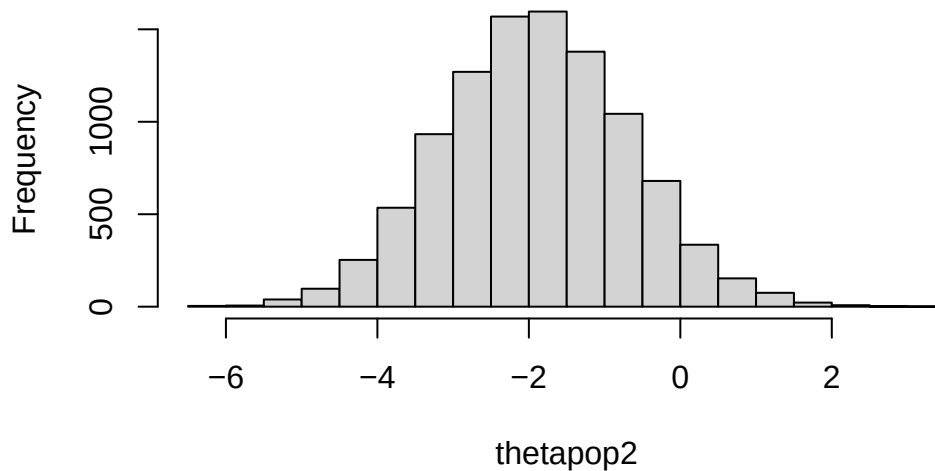
- (e) Create a sampling distribution of your weighted average incremental difference estimator. What size (n) should your repeated samples from the population be? What is the standard error for this estimator? Visually compare (with histograms) the sampling distribution to your bootstrap estimated sampling distribution for this estimator.

```
thetapop2<- rep(NA,S)

for( s in 1:S ){
  samind2 = sample(1:n,size=n,replace=T)
  dfsam2 = nba.pop[samind2,]
  p2<- length(dfsam2$draftlevel==2)/(length(dfsam2$draftlevel==2) + length(dfsam2$draftlevel
  p3<- length(dfsam2$draftlevel==3)/(length(dfsam2$draftlevel==2) + length(dfsam2$draftlevel
  thetapop2[s] = (-p2)*mean(dfsam2$net_rating[dfsam2$draftlevel==1]) + (p2-p3)*mean(dfsam2$net_rating[dfsam2$draftlevel==2]) + p3*mean(dfsam2$net_rating[dfsam2$draftlevel==3])
}

hist(thetapop2)
```

Histogram of thetapop2



```
sd(thetapop2)
```

```
[1] 1.224264
```

More normal and tighter than the estimated, our estimated standard error was higher.

- (f) What is the least squares difference in mean net rating across draft levels in the population?

$$\sum_x \tilde{w}(x)\mu(x)$$

where $\tilde{w}(x) = \frac{m_x(x-\bar{X})}{\sum_i (X_i - \bar{X})^2}$ How does this compare to your estimate?

```
xvarpop<- rep(NA, length(nba.pop$net_rating))
for(i in 1:length(nba.pop$net_rating)){
  xvarpop[i]<- (nba.pop$draftlevel[i] - mean(nba.pop$draftlevel))^2
}
denompop<- sum(xvarpop)
```

```
((np1*(1-mean(nba.pop$draftlevel)))/denompop)*mu1 + ((np2*(2-mean(nba.pop$draftlevel)))/de
```

```
[1] -2.201974
```

```
#Just to check because it should be the same as the slope:
modpop<- lm(net_rating~draftlevel, data=nba.pop)
summary(modpop)
```

Call:

```
lm(formula = net_rating ~ draftlevel, data = nba.pop)
```

Residuals:

Min	1Q	Median	3Q	Max
-247.866	-3.566	2.434	7.136	139.838

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.06776	2.01925	0.034	0.9732
draftlevel	-2.20197	0.99473	-2.214	0.0272 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.99 on 603 degrees of freedom

Multiple R-squared: 0.008061, Adjusted R-squared: 0.006416

F-statistic: 4.9 on 1 and 603 DF, p-value: 0.02723

The true least squares average increment is higher (closer to 0).

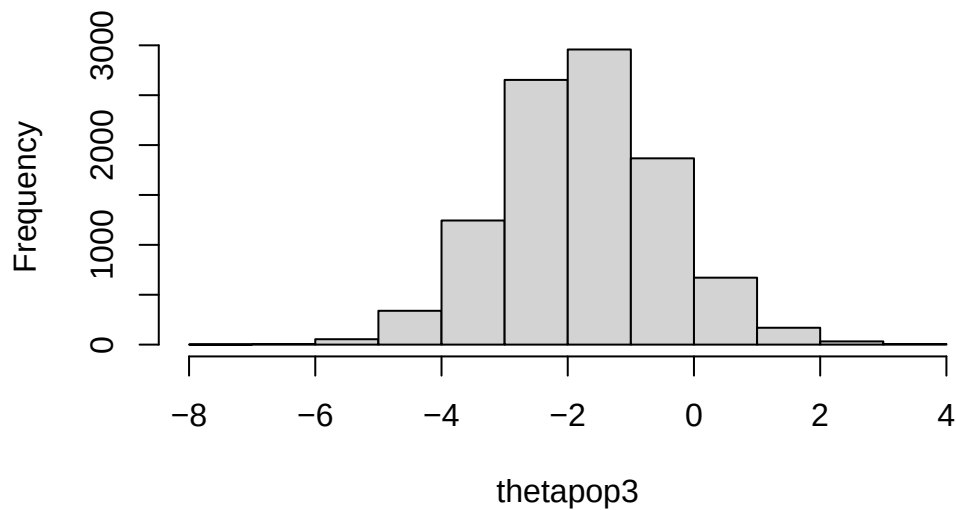
- (g) Create a sampling distribution of your least squares estimator. What size (n) should your repeated samples from the population be? What is the standard error for this estimator? Visually compare (with histograms) the sampling distribution to your bootstrap estimated sampling distribution for this estimator.

```
thetapop3<- rep(NA,S)

for( s in 1:S ){
  samind3 = sample(1:n,size=n,replace=T)
  dfsam3 = nba.pop[samind3,]
  ols<- lm(dfsam3$net_rating~ dfsam3$draftlevel)
  thetapop3[s] = ols$coef[2]
}

hist(thetapop3)
```

Histogram of thetapop3



```
sd(thetapop3)
```

```
[1] 1.307645
```

Sampling distribution is more normal and much tighter than estimated sampling distribution, standard error lower than estimated standard error.

- (h) Overall, how did your estimates compare to the population? How similar were your estimated standard errors to the standard errors of your estimators? Why might the estimators have different standard errors?

The subpopulations have pretty high variance and so do the subsamples. All of our estimators put a lot of weight on the undrafted players and there is the subsample (and subpopulation) with the greatest conditional variance. Our bootstrap estimated sampling distributions are not as similar to the actual sampling distributions (that we generated using the population) as we might want. Our sample is pretty small. All of our CIs covered the population estimands; our estimated standard errors were all larger than the standard errors of our estimators, but at least that means we are being more conservative.